

Eigenbases

Defⁿ $T: V \longrightarrow V$
F-linear transformation

An eigenbasis for T

is a basis for V consisting of
eigenvectors for T

Observation (i)

T is diagonalizable

$\iff$ there exists an
eigenbasis for T

Proposition ①

Suppose that $\{v_1, v_2, \dots, v_m\} \subseteq V$
is a set of ^{non-zero} eigenvectors for T
with distinct eigenvalues

i.e. $\cdot T(v_i) = \lambda_i v_i, \lambda_i \in F$

$\cdot \lambda_i \neq \lambda_j \text{ when } i \neq j$

Then $\{v_1, v_2, \dots, v_m\}$ is a linearly independent set

Pf: If they were not linearly independent, we could (after reindexing) write

$$\lambda_i \cdot v_i = \lambda_i [a_2 v_2 + a_3 v_3 + \dots + a_m v_m]$$

one of these is non-zero
($T(\cdot)$)

$$T(v_1) = a_2 T(v_2) + a_3 T(v_3) + \dots + a_m T(v_m)$$

$$\lambda_1 v_1 = a_2 \lambda_2 v_2 + a_3 \lambda_3 v_3 + \dots + a_m \lambda_m v_m$$

$$a_2 \lambda_1 v_2 + a_3 \lambda_1 v_3 + \dots + a_m \lambda_1 v_m$$

$$a_2(\lambda_1 - \lambda_2)v_2 + a_3(\lambda_1 - \lambda_3)v_3 + \dots + a_m(\lambda_1 - \lambda_m)v_m$$

one of these is non-zero

$\Rightarrow \{v_2, \dots, v_m\}$ are linearly dependent